

Credit Risk Prediction Using Machine Learning Techniques in R

Abstract

Credit risk assessment is a crucial task for financial institutions in order to minimize financial losses and maintain stable lending operations. The purpose of this study is to develop a credit risk prediction model using machine learning techniques implemented in the R programming language. The German Credit Dataset was used to analyze the financial characteristics of loan applicants. Three classification models—Logistic Regression, Decision Tree, and Random Forest—were applied to predict whether an applicant represents a good or bad credit risk. The performance of the models was evaluated using accuracy, precision, recall, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). The results indicate that the Random Forest model provides the highest predictive performance. The findings suggest that machine learning techniques can effectively support financial institutions in improving credit evaluation processes.

1. Introduction

Credit risk refers to the possibility that a borrower will fail to meet their debt obligations in accordance with agreed contractual terms. Effective credit risk assessment is essential for banks and financial institutions in order to prevent financial losses and maintain a stable financial system.

Traditionally, credit risk evaluation has been performed using statistical models and manual financial analysis. However, with the rapid development of data science and the increasing availability of financial data, machine learning techniques have emerged as powerful tools for predictive modeling.

Machine learning algorithms allow financial institutions to analyze complex patterns in historical loan data and identify borrowers who are more likely to default on their loans.

The objective of this study is to develop a predictive credit risk model using machine learning techniques in the R programming environment. The analysis focuses on comparing different classification algorithms and evaluating their effectiveness in predicting credit risk.

2. Dataset Description

The dataset used in this study is the **German Credit Dataset**, which is widely used in credit risk modeling research.

The dataset contains financial and demographic information about loan applicants. The goal is to predict whether an applicant represents a good credit risk or a bad credit risk.

The dataset includes the following characteristics:

- Number of observations: **1000**

- Number of explanatory variables: **20**
- Target variable: **Credit Risk Classification**

The target variable divides applicants into two categories:

- **Good credit risk**
- **Bad credit risk**

Several variables describing the financial situation of the borrower are included in the dataset.

Examples of variables include:

Variable	Description
Age	Age of the loan applicant
Credit Amount	Amount of credit requested
Duration	Duration of the loan
Employment	Employment status
Savings Account	Savings account balance
Checking Account	Checking account status
Housing	Housing ownership status

These variables provide valuable information that can be used to predict loan repayment behavior.

3. Methodology

This study follows a structured machine learning pipeline consisting of several stages including data preprocessing, exploratory data analysis, model development, and model evaluation.

3.1 Data Preprocessing

Before implementing machine learning algorithms, the dataset was prepared through several preprocessing steps.

First, the dataset was inspected in order to understand its structure and identify potential issues such as missing values or incorrect data types.

Categorical variables were encoded into numerical representations in order to make them suitable for machine learning models. Additionally, the dataset was divided into training and testing subsets.

The dataset was split as follows:

- **Training dataset: 80 % of the observations**

- **Testing dataset: 20% of the observations**

The training dataset was used to train the machine learning models, while the testing dataset was used to evaluate their predictive performance.

3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted in order to gain insights into the structure of the dataset and identify patterns within the data.

Several visualizations were generated to explore the distribution of variables and their relationships with credit risk.

The following analyses were conducted:

- Histogram analysis for numerical variables such as age, loan amount, and loan duration
- Boxplot analysis comparing financial variables across credit risk categories
- Correlation analysis for numerical variables

The correlation matrix was used to analyze the relationships between numerical variables in the dataset. This analysis helps identify potential dependencies among financial indicators.

3.3 Machine Learning Models

Three classification models were implemented in this study to predict credit risk.

Logistic Regression

Logistic regression is a widely used statistical method for binary classification problems. It models the probability that a given observation belongs to a particular class.

Logistic regression is commonly used in credit risk modeling due to its simplicity and interpretability. The model provides probability estimates that can be interpreted as the likelihood of default.

Decision Tree

Decision tree models classify observations by recursively splitting the dataset according to feature values.

Decision trees are particularly useful for credit risk analysis because they provide transparent and interpretable decision rules. Financial institutions often prefer interpretable models when making lending decisions.

Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions.

This approach improves predictive performance by reducing overfitting and capturing complex nonlinear relationships within the data.

Random Forest models are widely used in credit risk prediction due to their high predictive accuracy.

4. Model Evaluation

In order to evaluate the predictive performance of the models, several evaluation metrics were used.

Accuracy

Accuracy measures the proportion of correctly classified observations among all predictions.

Precision

Precision measures the proportion of correctly predicted positive cases among all predicted positive cases.

Recall

Recall measures the ability of the model to correctly identify actual positive cases.

ROC Curve and AUC

The Receiver Operating Characteristic (ROC) curve illustrates the relationship between the true positive rate and false positive rate.

The Area Under the Curve (AUC) provides a summary measure of model performance. Higher AUC values indicate better predictive capability.

5. Results

The performance of the three machine learning models was compared using the evaluation metrics described above.

The analysis showed that:

- Logistic Regression provided a strong baseline model.
- Decision Trees offered interpretable classification rules but slightly lower predictive performance.
- Random Forest achieved the highest accuracy and AUC values among the models tested.

The Random Forest model also identified the most influential variables affecting credit risk.

Important predictors include:

- Credit amount
- Loan duration
- Age
- Account status

These variables have a significant impact on the probability of loan default.

6. Discussion

The results of this study demonstrate that machine learning models can effectively classify loan applicants based on their credit risk.

Among the models tested, Random Forest showed the strongest predictive performance due to its ability to capture nonlinear relationships and interactions between variables.

Although logistic regression is simpler, it remains valuable due to its interpretability and statistical foundation.

Decision trees provide transparent decision-making structures that can be easily interpreted by financial analysts and decision-makers.

Therefore, the choice of model may depend on the trade-off between predictive accuracy and interpretability.

7. Conclusion

This study investigated the application of machine learning techniques to credit risk prediction using the R programming language.

Three classification models—Logistic Regression, Decision Tree, and Random Forest—were implemented and evaluated using the German Credit Dataset.

The results indicate that Random Forest provides the highest predictive performance among the models considered in this study.

The findings highlight the potential of machine learning methods to enhance credit risk assessment and support financial institutions in making more informed lending decisions.

Future research may focus on incorporating larger datasets, additional financial indicators, and advanced machine learning algorithms in order to further improve predictive accuracy.